■■ Insurance Customer Risk & Claims Analytics Project

Data Analyst — Python + MySQL + Power BI + Simple Streamlit

GROUP MEMBERS:

1. _____
2. _____
3. _____

■ Scenario (Realistic Business Context)

A leading insurance company provides policies to customers across different cities, age groups, occupations and insurance categories.

The company wants to analyze:

- Customer demographics
- Insurance policy performance
- Premium collection
- Claim patterns
- Claim approval/rejection
- Customer risk
- Policy performance
- Customer retention

You have been hired as a **Data Analyst** to:

- Analyze insurance customer data
- Identify high-risk customers and policies
- Analyze claim patterns
- Identify factors affecting claim amounts
- Analyze premium and revenue trends
- Build an interactive Power BI dashboard
- Use Python for EDA and analysis
- Use MySQL for data storage and business queries

■ 1. Generate / Collect Realistic Insurance Data

Create a dataset containing **50,000+ insurance customer/policy records**.

Customer Information

- Customer_ID
- Customer_Name

- Age
- Gender
- Marital_Status
- Occupation
- City
- Income

Policy Information

- Policy_ID
- Policy_Type (Health, Life, Motor, Home, Travel)
- Policy_Start_Date
- Policy_End_Date
- Policy_Status (Active, Expired, Cancelled)
- Coverage_Amount
- Premium_Amount
- Payment_Frequency

Claim Information

- Claim_ID
- Claim_Date
- Claim_Amount
- Claim_Status (Approved, Rejected, Pending)
- Claim_Type
- Claim_Processing_Days

Risk Information

- Risk_Level (Low, Medium, High)
- Previous_Claims
- Customer_Risk_Score

Target Variable: Claim_Status / Claim_Amount OR Risk_Level

■■ 2. MySQL Integration

Use **MySQL** for storing and analyzing the insurance data.

Create Tables

- Customers
- Policies
- Claims
- Insurance_Transactions

SQL Tasks

- Insert insurance records
- Retrieve customer information
- Analyze policy performance
- Analyze claim performance
- Find high-risk customers
- Calculate premium and claim metrics
- Identify profitable and loss-making policy types

SQL Concepts

- SELECT
- WHERE
- GROUP BY
- ORDER BY
- Aggregate Functions
- Joins
- Subqueries
- CTEs
- Window Functions
- Views

■ 3. EDA + Feature Engineering

Using: **Pandas, Matplotlib, Seaborn**

Customer Analysis

- Total Customers
- Customers by Age Group
- Customers by Gender
- Customers by Occupation
- Customers by City
- Income Distribution

Policy Analysis

- Total Policies
- Active vs Expired Policies
- Policies by Type
- Premium by Policy Type
- Coverage Amount Analysis

Claims Analysis

- Total Claims
- Approved Claims
- Rejected Claims
- Pending Claims
- Average Claim Amount
- Claim Amount by Policy Type

Risk Analysis

- High-Risk Customers
- Medium-Risk Customers
- Low-Risk Customers
- Risk by Age Group
- Risk by Policy Type
- Risk by Previous Claims

Feature Engineering

- Customer Risk Score
- Claim Frequency
- Claim-to-Premium Ratio
- Policy Value Score
- Customer Lifetime Value
- Policy Profitability Score

■ 4. Machine Learning / Predictive Analysis

Problem 1 – Claim Amount Prediction

Predict: **Expected Claim Amount**

Algorithms: Linear Regression, Random Forest Regressor, XGBoost Regressor

Metrics: MAE, RMSE, R^2 Score

Problem 2 – Risk Classification

Predict: **Customer Risk Level**

Algorithms: Logistic Regression, Random Forest, XGBoost

Metrics: Accuracy, Precision, Recall, F1 Score

Hyperparameter Tuning: GridSearchCV, RandomizedSearchCV

Save final model using: Joblib

■ 5. Power BI Dashboard

Page 1 – Executive Overview

KPIs:

- Total Customers
- Total Policies
- Total Premium
- Total Claims
- Average Claim Amount
- Claim Approval Rate

Charts: Customer Trend, Premium Trend, Claims Trend, Revenue by Policy Type

Page 2 – Customer Analytics

Charts: Customers by Age Group, Customers by Gender, Customers by Occupation, Customers by City, Income Distribution

Page 3 – Policy Analytics

Charts: Policies by Type, Active vs Expired Policies, Premium by Policy Type, Coverage Amount by Policy Type, Policy Performance

Page 4 – Claims Analytics

Charts: Claims Trend, Approved vs Rejected Claims, Claim Amount by Policy Type, Average Claim Amount, Claim Processing Time

Page 5 – Risk Analytics

Charts: High / Medium / Low Risk Customers, Risk by Age Group, Risk by Policy Type, Top High-Risk Customers, Claim Frequency vs Risk

■ 6. Streamlit Frontend Application

Streamlit should be kept **simple**, because the main interactive dashboard will be created in Power BI.

1. ■ Overview

- Total Customers
- Total Policies
- Total Premium
- Total Claims
- Average Claim Amount
- Simple charts for Policy Type, Claim Status and Risk Level

2. ■ Customer Explorer

- Search/filter by Customer ID, City, Policy Type and Risk Level
- Display Customer Details, Policy Details, Premium Amount, Claim History and Risk Level

3. ■ Risk Prediction

Input: Age, Income, Policy Type, Previous Claims, Premium Amount, Coverage Amount

Output: Predicted Risk Level and Risk Score

Suggested actions: High Risk → Review customer profile; Medium Risk → Monitor claims; Low Risk → Standard policy processing

4. ■ Analytics

Display simple charts for Customer Analysis, Policy Analysis, Claim Analysis and Risk Analysis.

Filters: Policy Type, City, Risk Level and Date Range

■ 7. Insurance Reports

- Daily Insurance Report
- Weekly Insurance Report
- High-Risk Customer Report
- Claims Summary Report
- Policy Performance Report
- Premium Analysis Report

■ Final Deliverables

```
Insurance_Customer_Risk_Analytics/
■
■■■ data/
■   ■■■ raw_insurance_data.csv
■   ■■■ cleaned_insurance_data.csv
■
■■■ notebooks/
■   ■■■ EDA.ipynb
■   ■■■ Feature_Engineering.ipynb
■   ■■■ ML_Model.ipynb
■
■■■ sql/
■   ■■■ schema.sql
■   ■■■ queries.sql
■
■■■ models/
■   ■■■ risk_prediction.pkl
■   ■■■ claim_prediction.pkl
■
■■■ powerbi/
■   ■■■ Insurance_Analytics.pbix
■
■■■ streamlit_app/
■   ■■■ app.py
■
■■■ reports/
■   ■■■ Daily_Insurance_Report.csv
■   ■■■ Weekly_Insurance_Report.csv
■   ■■■ Claims_Report.csv
■   ■■■ High_Risk_Customer_Report.csv
■
■■■ README.md
```

■ Project Coverage

- Data Collection
- Data Cleaning

- Python EDA
- Feature Engineering
- MySQL
- SQL Business Analysis
- Machine Learning
- Power BI Dashboard
- Simple Streamlit Application
- Insurance Risk Analysis
- Claims Analysis
- Business Reporting
- Enterprise-Level Project Documentation